

CHAOYUAN LIN

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Education

University of California, San Diego

La Jolla, CA

Major: Mathematics-Computer Science, Bachelor's of Science

June 2026

Courses: Data Structures, Algorithms, Operating Systems, Computer Networks, Deep Learning, Machine Learning, Database

Projects

NBA Dashboard | *Docker, React.js, Spring Boot, PostgreSQL, Tailwind CSS, Data Analysis*

September 2025

- Built a fullstack NBA analytics web application that was deployed in containerized environments, enhancing **scalability** and **ease of deployment**, by implementing a React frontend with **interactive data visualizations** and a Spring Boot backend exposing **RESTful APIs**, with Docker.
- Processed and cleaned **18,000+** player records covering multiple NBA seasons to ensure reliable analysis and insights, by designing a backend **data pipeline** that performed **league filtering**, **validated player positions**, and **calculated per-game statistics**.
- Enabled users to explore **46+ NBA seasons** with insightful statistic data, by transforming historical player records into a structured dataset that supports **seasonal comparisons**, **player evaluations**, and **team analysis**.

PeerFS | *Go, TCP, P2P Networking, Encryption, Distributed Systems*

August 2025

- Built a distributed file storage system in Go that replicated files across a **peer-to-peer**, improving **fault tolerance** and **data availability** by enabling recovery of files from peers even after local deletion, through **content-addressable storage** and **automatic replication**.
- Benchmarked storage path on **100** file batches, achieving **137 MB/s** write throughput for **10 MB** objects and **109 MB/s** for **1 MB** objects, demonstrating efficient **streaming I/O** and **low per-file overhead** at larger sizes.
- Designed a custom **TCP-based protocol** enabling reliable peer discovery, messaging, and file replication across nodes as well as developed **message serialization** and peer handshake logic to ensure **secure communication** and **consistent replication** throughout the distributed network.

TritonTube | *Backend, Distributed Systems, Go, gRPC, etcd, RocksDB, Consistent Hashing*

June 2025

- Engineered a distributed video streaming platform using **consistent hashing** for **horizontal scaling** and **automatic data rebalancing** across storage nodes, with **gRPC-based communication** and **RocksDB-backed storage**.
- Designed a **fault-tolerant metadata service** using a 3-node **etcd cluster**, ensuring strong consistency and high availability under node failures.
- Reduced video upload latency by **97%** for 475 MPEG-DASH segments and optimized migration by **98%** with **gRPC based batch transfers**.

Work Experience

Math Tutor

August 2022 - December 2022

- Supported **50+** students in mastering core math concepts by adapting explanations with graphs and examples to strengthen student problem solving skills.

Certificate

AWS Cloud Technical Essentials | *Link*

September 2025

- Gained practical experience with AWS compute, storage, and networking services including **EC2**, **S3**, **DynamoDB**, and **VPC**, with a focus on **IAM** and **security** best practices.
- Configured scalable and highly available architectures by implementing **Auto Scaling Groups**, **Elastic Load Balancing**, and **CloudWatch monitoring**.

Skills

Languages: Java, GoLang, Python, C/C++, SQL

Tools: Git/GitHub, Docker, Linux, Postman, VS Code

Frameworks/Technologies: React.js, PyTorch, gRPC, Spring Boot, etcd, PostgreSQL